

Tables

Table 1. Sample sizes and counts of unique families, genera, and species for each survey era and sea combination. The historical survey is the Smithsonian survey that sampled in 1978 and 1979 (SI 1978/79) and the modern survey is the Silliman University survey that duplicated the same stations in 2019 and 2022 (SU 2019/2022). The number of genera include taxa that were identified to the family level and were combined into a species complex. The number of species include taxa that were identified to the genus or family level and were combined into a species complex.

Era	Sea	Stations	Families	Genera	Species
All	All	42	77	292	957
Historical	All	21	59	237	677
Modern	All	21	62	226	617
All	Bohol Sea	26	69	231	670
All	Sulu Sea	16	70	240	667
Historical	Bohol Sea	13	52	186	460
Modern	Bohol Sea	13	56	174	396
Historical	Sulu Sea	8	52	186	456
Modern	Sulu Sea	8	55	178	410

Table 2. Summary of alpha diversity indices for reef fish communities sampled in the Sulu and Bohol Seas during two survey eras (Historical: SI 1978/1979 and Modern: SU 2019/2022). Values are presented as the mean and standard deviation (SD) for Shannon diversity (H), Simpson’s diversity, inverse Simpson’s diversity, unbiased Simpson’s diversity, and Fisher’s alpha, calculated across all stations by survey era, by sea, and by era:sea combination. “All” indicates values pooled across both eras or seas.

Era	Sea	Mean Shannon's H (SD)	Mean Simpson Index (SD)	Mean Inverse Simpson (SD)	Mean Unbiased Simpson (SD)	Mean Alpha (SD)
Historical	All	3.352 (0.449)	0.921 (0.049)	16.581 (7.306)	0.923 (0.049)	25.681 (8.407)
Modern	All	3.225 (0.508)	0.894 (0.102)	14.956 (7.380)	0.897 (0.103)	26.806 (6.580)
All	Bohol Sea	3.237 (0.421)	0.910 (0.054)	14.919 (7.208)	0.913 (0.054)	24.547 (7.688)
All	Sulu Sea	3.371 (0.563)	0.904 (0.113)	17.150 (7.468)	0.905 (0.113)	29.000 (6.423)
Historical	Bohol Sea	3.203 (0.488)	0.906 (0.056)	14.588 (7.755)	0.908 (0.056)	22.811 (8.254)
Modern	Bohol Sea	3.272 (0.359)	0.914 (0.054)	15.249 (6.917)	0.918 (0.054)	26.284 (6.962)
Historical	Sulu Sea	3.594 (0.246)	0.946 (0.015)	19.820 (5.481)	0.947 (0.015)	30.346 (6.705)
Modern	Sulu Sea	3.149 (0.712)	0.861 (0.151)	14.480 (8.554)	0.863 (0.152)	27.654 (6.269)

Table 3. Results of linear mixed-effect models (LME) testing spatiotemporal differences on alpha diversity metrics for paired reef fish communities. Each metric was analyzed using a linear mixed-effect model of the form: $\text{metric} \sim \text{Era} * \text{Sea} + (1 \mid \text{station_pair})$ where Era (SI 1978/1979 vs. SU 2019/2022) and Sea (Bohol Sea vs. Sulu Sea) were treated as fixed effects and station_pair (duplicated site code) as a random effect (accounting for repeated sampling). For each metric, fixed-effect estimates are shown with standard errors (SE) and 95% confidence intervals (CI). Analysis of Deviance (Type III ANOVA) tests are shown with Wald chi-square statistics (Chi-Sq) and corresponding p-values.

Metric	Term	Fixed Effects			ANOVA	
		Estimate	SE	CI	Chi-Sq	p
Shannon's H	(Intercept)	3.20	0.13	2.95 - 3.46	606	0.000
Shannon's H	Era	0.07	0.16	-0.24 - 0.38	0.19	0.661
Shannon's H	Sea	0.39	0.21	-0.02 - 0.8	3.44	0.064
Shannon's H	Era:Sea	-0.51	0.26	-1.01 - -0.01	4.05	0.044
Simpson Index	(Intercept)	0.91	0.02	0.86 - 0.95	1739	0.000
Simpson Index	Era	0.01	0.03	-0.05 - 0.07	0.07	0.795
Simpson Index	Sea	0.04	0.04	-0.03 - 0.11	1.29	0.255
Simpson Index	Era:Sea	-0.09	0.05	-0.19 - 0	3.67	0.055
Inverse Simpson	(Intercept)	14.59	2.02	10.63 - 18.55	52	0.000
Inverse Simpson	Era	0.66	2.46	-4.17 - 5.49	0.07	0.788
Inverse Simpson	Sea	5.23	3.27	-1.19 - 11.65	2.55	0.110
Inverse Simpson	Era:Sea	-6.00	3.99	-13.83 - 1.82	2.26	0.133
Unbiased Simpson	(Intercept)	0.91	0.02	0.87 - 0.95	1720	0.000
Unbiased Simpson	Era	0.01	0.03	-0.05 - 0.07	0.11	0.735
Unbiased Simpson	Sea	0.04	0.04	-0.03 - 0.11	1.22	0.269
Unbiased Simpson	Era:Sea	-0.09	0.05	-0.19 - 0	3.74	0.053
Alpha	(Intercept)	22.81	2.01	18.88 - 26.74	129	0.000
Alpha	Era	3.47	2.24	-0.92 - 7.87	2.40	0.121
Alpha	Sea	7.54	3.25	1.16 - 13.91	5.37	0.020
Alpha	Era:Sea	-6.17	3.63	-13.28 - 0.95	2.88	0.089

Table 4. PERMANOVA of community centroids by survey era, sea, and the era:sea interaction on Bray–Curtis distances (bc_dist) using the *adonis2* function from the *vegan* package with the formula *bc_dist ~ study * sea* and 999 permutations. df: degrees of freedom; Sum Sq: sum of squares; Mean Sq: mean square; F-statistic: pseudo-F statistic; R²: proportion of variance explained; *p*: p-value.

Factor	df	Sum Sq	Mean Sq	F-statistic	R ²	<i>p</i>
Era	1	1.023	1.023	2.531	0.057	0.001
Sea	1	0.881	0.881	2.178	0.049	0.002
Era:Sea	1	0.578	0.578	1.429	0.032	0.015
Residual	38	15.366	0.404		0.861	
Total	41	17.848			1.000	

Table S1. Temporal change in alpha diversity indices means for 21 duplicated reef fish community stations in the Sulu and Bohol Seas sampled during two survey eras (SI 1978/1979 and SU 2019/2022). Values are presented as the difference in the mean for Shannon diversity (H), Simpson's diversity, inverse Simpson's diversity, unbiased Simpson's diversity, and Fisher's alpha, calculated at each station and each sea. "All" indicates values pooled across both seas.

Station Code	Sea	Province	Municipality	Δ Shannon's H	Δ Simpson	Δ Inverse Simpson	Δ Unbiased Simpson	Δ Alpha
All	Bohol Sea	All	All	0.069	0.008	0.662	0.010	3.473
All	Sulu Sea	Palawan	All	-0.445	-0.085	-5.340	-0.084	-2.693
01_bohol	Bohol Sea	Negros Oriental	Siaton	-0.005	-0.001	-0.229	0.002	2.971
02_bohol	Bohol Sea	Negros Oriental	Siaton	0.235	0.080	9.080	0.088	1.979
03_bohol	Bohol Sea	Negros Oriental	Siaton	-0.429	-0.132	-14.550	-0.132	15.715
04_bohol	Bohol Sea	Negros Oriental	Siaton	0.467	0.074	4.438	0.075	8.167
05_bohol	Bohol Sea	Negros Oriental	Siaton	0.001	-0.001	-1.045	-0.002	-1.240
06_bohol	Bohol Sea	Negros Oriental	Siaton	-1.181	-0.152	-16.183	-0.150	-15.650
07_bohol	Bohol Sea	Negros Oriental	Siaton	0.697	0.098	13.192	0.103	5.757
08_bohol	Bohol Sea	Siquijor	Larena	0.412	0.034	8.238	0.037	10.847
09_bohol	Bohol Sea	Siquijor	San Juan	0.749	0.075	13.719	0.074	10.158
10_bohol	Bohol Sea	Siquijor	San Juan	0.034	-0.009	-4.561	-0.009	7.684
11_bohol	Bohol Sea	Siquijor	San Juan	-0.153	-0.010	-3.002	-0.007	-0.600
12_bohol	Bohol Sea	Siquijor	Siquijor	-0.579	-0.047	-4.705	-0.042	-5.425
13_bohol	Bohol Sea	Siquijor	Siquijor	0.650	0.092	4.211	0.095	4.786
14_sulu	Sulu Sea	Palawan	Cuyo	-0.233	-0.006	-2.144	-0.006	-8.411
15_sulu	Sulu Sea	Palawan	Cuyo	-0.513	-0.047	-17.612	-0.045	-8.490
16_sulu	Sulu Sea	Palawan	Cuyo	-0.241	0.001	0.346	0.002	-9.525
17_sulu	Sulu Sea	Palawan	Magsaysay	0.099	0.003	1.624	0.004	3.149
18_sulu	Sulu Sea	Palawan	Magsaysay	-0.900	-0.283	-15.102	-0.284	13.467
19_sulu	Sulu Sea	Palawan	Magsaysay	-0.044	-0.010	-1.694	-0.010	0.880
20_sulu	Sulu Sea	Palawan	Magsaysay	-1.648	-0.341	-10.205	-0.343	-8.708
21_sulu	Sulu Sea	Palawan	Magsaysay	-0.077	0.004	2.067	0.008	-3.904

Table S2. Pairwise post-hoc comparisons (Tukey-adjusted) for estimated marginal means from linear mixed-effects models of alpha diversity indices. Contrasts include all six pairwise comparisons between the four Era:Sea combinations. Refer to Table 3 for details.

Metric	Contrast	Mean Difference	SE	df	t-ratio	<i>p</i>
Shannon's H	historical bohol - modern bohol	-0.069	0.157	19	-0.439	0.971
Shannon's H	historical bohol - historical sulu	-0.391	0.211	35	-1.855	0.266
Shannon's H	historical bohol - modern sulu	0.054	0.211	35	0.254	0.994
Shannon's H	modern bohol - historical sulu	-0.322	0.211	35	-1.527	0.432
Shannon's H	modern bohol - modern sulu	0.123	0.211	35	0.581	0.937
Shannon's H	historical sulu - modern sulu	0.445	0.201	19	2.215	0.155
Simpson	historical bohol - modern bohol	-0.008	0.030	19	-0.260	0.994
Simpson	historical bohol - historical sulu	-0.040	0.035	38	-1.138	0.669
Simpson	historical bohol - modern sulu	0.045	0.035	38	1.274	0.585
Simpson	modern bohol - historical sulu	-0.032	0.035	38	-0.917	0.796
Simpson	modern bohol - modern sulu	0.053	0.035	38	1.495	0.450
Simpson	historical sulu - modern sulu	0.085	0.038	19	2.232	0.150
Inverse Simpson	historical bohol - modern bohol	-0.662	2.464	19	-0.269	0.993
Inverse Simpson	historical bohol - historical sulu	-5.232	3.275	36	-1.598	0.393
Inverse Simpson	historical bohol - modern sulu	0.108	3.275	36	0.033	1.000
Inverse Simpson	modern bohol - historical sulu	-4.570	3.275	36	-1.396	0.510
Inverse Simpson	modern bohol - modern sulu	0.770	3.275	36	0.235	0.995
Inverse Simpson	historical sulu - modern sulu	5.340	3.141	19	1.700	0.351
Unbiased Simpson	historical bohol - modern bohol	-0.010	0.030	19	-0.338	0.986
Unbiased Simpson	historical bohol - historical sulu	-0.039	0.035	38	-1.106	0.688
Unbiased Simpson	historical bohol - modern sulu	0.045	0.035	38	1.267	0.589
Unbiased Simpson	modern bohol - historical sulu	-0.029	0.035	38	-0.819	0.845
Unbiased Simpson	modern bohol - modern sulu	0.055	0.035	38	1.554	0.417
Unbiased Simpson	historical sulu - modern sulu	0.084	0.038	19	2.193	0.161

Alpha	historical bohol - modern bohol	-3.473	2.241	19	-1.550	0.429
Alpha	historical bohol - historical sulu	-7.536	3.251	33	-2.318	0.114
Alpha	historical bohol - modern sulu	-4.843	3.251	33	-1.490	0.455
Alpha	modern bohol - historical sulu	-4.063	3.251	33	-1.250	0.600
Alpha	modern bohol - modern sulu	-1.370	3.251	33	-0.421	0.974
Alpha	historical sulu - modern sulu	2.693	2.857	19	0.943	0.783

Table S3. Results from a two-way repeated measures ANOVA test for homogeneity of dispersions (*betadisper* function in the *vegan* package) on Bray–Curtis distances using the formula *distance ~ study * sea* to test for effects of survey era, sea, and their interaction (Era:Sea).

Factor	Df	Sum Sq	Mean Sq	F-statistic	<i>p</i>
Era	1	0.006	0.006	1.475	0.232
Sea	1	0.000	0.000	0.084	0.774
Era:Sea	1	0.005	0.005	1.363	0.250
Residuals	38	0.144	0.004		

Table S4. Results of linear mixed-effect models (LME) testing spatiotemporal differences on community-weighted mean trophic level (*cwm_troph*) for paired reef fish communities analyzed using a linear mixed-effect model of the form $cwm_troph \sim Era * Sea + (1 | station_pair)$ where Era (SI 1978/1979 vs. SU 2019/2022) and Sea (Bohol Sea vs. Sulu Sea) were treated as fixed effects and *station_pair* (duplicated site code) as a random effect (accounting for repeated sampling). Fixed-effect estimates are shown with standard errors (SE) and 95% confidence intervals (CI). Analysis of Deviance (Type III ANOVA) tests are shown with Wald chi-square statistics (Chi-Sq) and corresponding p-values.

Term	Fixed Effects			ANOVA	
	Estimate	SE	CI	Chi-Sq	<i>p</i>
(Intercept)	2.212	0.148	1.93 - 2.49	222	<2e-16
Era	-0.190	0.207	-0.59 - 0.21	0.838	0.360
Sea	0.138	0.240	-0.32 - 0.60	0.332	0.565
Era:Sea	-0.502	0.336	-1.15 - 0.15	2.232	0.135

Table S5. Bray–Curtis cluster with SIMPROF profile test ($\alpha = 0.05$) using the *simprof* function from the *clustsig* package. Seventeen SIMPROF significant clusters were identified from the 42 stations across the Bohol and Sulu Seas and the historical (SI 1978/79) and modern (SU 2019/22) eras surveyed. Refer to Figure 7 for the dendrogram.

Station Code	SIMPROF Significant Cluster	Era	Sea	Province	Municipality	Depth (m)
si_19_sulu	1	Historical	Sulu Sea	Palawan	Magsaysay	4.6
si_18_sulu	1	Historical	Sulu Sea	Palawan	Magsaysay	3.0
si_12_bohol	2	Historical	Bohol Sea	Siquijor	Siquijor	1.0
si_03_bohol	3	Historical	Bohol Sea	Negros Oriental	Siaton	3.0
si_01_bohol	3	Historical	Bohol Sea	Negros Oriental	Siaton	2.0
si_04_bohol	4	Historical	Bohol Sea	Negros Oriental	Siaton	3.0
su_19_sulu	4	Modern	Sulu Sea	Palawan	Magsaysay	4.6
su_20_sulu	5	Modern	Sulu Sea	Palawan	Magsaysay	13.6
su_14_sulu	6	Modern	Sulu Sea	Palawan	Cuyo	1.2
su_18_sulu	6	Modern	Sulu Sea	Palawan	Magsaysay	3.0
su_17_sulu	6	Modern	Sulu Sea	Palawan	Magsaysay	2.4
si_14_sulu	7	Historical	Sulu Sea	Palawan	Cuyo	1.0
si_17_sulu	8	Historical	Sulu Sea	Palawan	Magsaysay	2.0
si_20_sulu	9	Historical	Sulu Sea	Palawan	Magsaysay	13.7
si_15_sulu	9	Historical	Sulu Sea	Palawan	Cuyo	13.7
si_16_sulu	9	Historical	Sulu Sea	Palawan	Cuyo	17.4
si_13_bohol	10	Historical	Bohol Sea	Siquijor	Siquijor	30.5
si_08_bohol	11	Historical	Bohol Sea	Siquijor	Larena	30.5
si_21_sulu	11	Historical	Sulu Sea	Palawan	Magsaysay	21.3
su_16_sulu	12	Modern	Sulu Sea	Palawan	Cuyo	17.3
su_15_sulu	12	Modern	Sulu Sea	Palawan	Cuyo	13.6
su_10_bohol	13	Modern	Bohol Sea	Siquijor	San Juan	6.0
su_04_bohol	13	Modern	Bohol Sea	Negros Oriental	Siaton	3.0

su_13_bohol	14	Modern	Bohol Sea	Siquijor	Siquijor	30.0
su_11_bohol	14	Modern	Bohol Sea	Siquijor	San Juan	10.7
su_07_bohol	14	Modern	Bohol Sea	Negros Oriental	Siaton	18.3
su_06_bohol	14	Modern	Bohol Sea	Negros Oriental	Siaton	12.0
su_08_bohol	14	Modern	Bohol Sea	Siquijor	Larena	30.0
su_05_bohol	14	Modern	Bohol Sea	Negros Oriental	Siaton	9.1
su_21_sulu	14	Modern	Sulu Sea	Palawan	Magsaysay	21.2
su_03_bohol	15	Modern	Bohol Sea	Negros Oriental	Siaton	5.0
si_07_bohol	16	Historical	Bohol Sea	Negros Oriental	Siaton	18.3
si_11_bohol	16	Historical	Bohol Sea	Siquijor	San Juan	10.7
si_06_bohol	16	Historical	Bohol Sea	Negros Oriental	Siaton	12.2
si_10_bohol	16	Historical	Bohol Sea	Siquijor	San Juan	6.1
si_05_bohol	16	Historical	Bohol Sea	Negros Oriental	Siaton	9.1
su_01_bohol	17	Modern	Bohol Sea	Negros Oriental	Siaton	3.5
su_02_bohol	17	Modern	Bohol Sea	Negros Oriental	Siaton	2.5
su_09_bohol	17	Modern	Bohol Sea	Siquijor	San Juan	2.0
si_02_bohol	17	Historical	Bohol Sea	Negros Oriental	Siaton	2.0
si_09_bohol	17	Historical	Bohol Sea	Siquijor	San Juan	2.0
su_12_bohol	17	Modern	Bohol Sea	Siquijor	Siquijor	2.0

Table S6. Similarity percentages (*simper* function in the *vegan* package) of the top species that contribute to 20% of the cumulative differences between survey eras.

Species	Average	SD	Cumulative Sum	Historical	Modern	Ratio	<i>p</i>
<i>Verulux cypselurus</i>	0.025	0.112	0.026	0.0	101.9	0.222	0.997
<i>Plotosus lineatus</i>	0.022	0.052	0.050	11.3	7.6	0.432	0.885
<i>Chromis ternatensis</i>	0.022	0.065	0.073	31.2	1.7	0.336	0.028
<i>Ostorhinchus cyanosoma</i>	0.018	0.044	0.093	8.4	13.6	0.412	0.662
<i>Spratelloides gracilis</i>	0.018	0.078	0.111	0.0	29.9	0.229	0.991
<i>Spratelloides delicatulus</i>	0.018	0.057	0.130	11.0	12.0	0.311	0.221
<i>Pycnochromis retrofasciatus</i>	0.017	0.027	0.148	14.0	7.1	0.647	0.820
<i>Cirrhilabrus ryukyuensis</i>	0.014	0.024	0.163	0.0	14.7	0.572	0.947
<i>Ostorhinchus nigrofasciatus</i>	0.012	0.016	0.176	9.4	5.5	0.734	0.376
<i>Ostorhinchus multilineatus</i>	0.011	0.015	0.187	5.5	9.9	0.693	0.998
<i>Ostorhinchus fleurieu</i>	0.011	0.041	0.198	8.4	0.6	0.261	0.012
<i>Dascyllus reticulatus</i>	0.010	0.026	0.209	3.7	7.8	0.392	0.963

Table S7. Similarity percentages (*simper* function in the *vegan* package) of the top species that contribute to 20% of the cumulative differences between seas.

Species	Average	SD	Cumulative Sum	Bohol Sea	Sulu Sea	Ratio	<i>p</i>
<i>Verulux cypselurus</i>	0.035	0.137	0.037	0.0	133.7	0.257	0.001
<i>Spratelloides gracilis</i>	0.028	0.104	0.067	0.0	39.1	0.266	0.001
<i>Chromis ternatensis</i>	0.019	0.054	0.086	20.9	9.3	0.348	0.652
<i>Spratelloides delicatulus</i>	0.016	0.046	0.104	8.9	15.8	0.349	0.659
<i>Cirrhilabrus ryukyuensis</i>	0.016	0.030	0.121	3.5	13.6	0.528	0.116
<i>Plotosus lineatus</i>	0.015	0.038	0.137	14.9	0.6	0.399	1.000
<i>Ostorhinchus cyanosoma</i>	0.015	0.038	0.153	12.6	8.4	0.395	0.980
<i>Pycnochromis retrofasciatus</i>	0.014	0.023	0.167	14.9	3.6	0.607	0.999
<i>Chrysiptera rollandi</i>	0.013	0.016	0.181	1.7	12.6	0.816	0.001
<i>Enneapterygius</i>	0.013	0.034	0.194	0.5	20.5	0.365	0.002
<i>Cirrhilabrus</i>	0.012	0.030	0.207	1.0	11.5	0.385	0.009

Table S8. Similarity percentages (*simper* function in the *vegan* package) of the top species that contribute to 20% of the cumulative differences between survey eras in the Bohol Sea.

Species	Average	SD	Cumulative Sum	Historical Bohol Sea	Modern Bohol Sea	Ratio	<i>p</i>
<i>Plotosus lineatus</i>	0.039	0.068	0.042	18.2	11.6	0.579	0.006
<i>Chromis ternatensis</i>	0.027	0.080	0.071	39.8	2.0	0.334	0.374
<i>Pycnochromis retrofasciatus</i>	0.026	0.031	0.098	18.6	11.2	0.843	0.011
<i>Ostorhinchus cyanosoma</i>	0.025	0.059	0.125	6.0	19.2	0.421	0.290
<i>Spratelloides delicatulus</i>	0.022	0.074	0.149	17.8	0.0	0.302	0.400
<i>Ostorhinchus fleurieu</i>	0.020	0.056	0.170	13.5	0.6	0.349	0.001
<i>Ostorhinchus nigrofasciatus</i>	0.019	0.020	0.190	14.1	7.6	0.964	0.001
<i>Pycnochromis atripes</i>	0.018	0.050	0.209	26.6	0.0	0.356	0.001

Table S9. Similarity percentages (*simper* function in the *vegan* package) of the top species that contribute to 20% of the cumulative differences between survey eras in the Sulu Sea.

Species	Average	SD	Cumulative Sum	Historical Sulu Sea	Modern Sulu Sea	Ratio	<i>p</i>
<i>Verulux cypselurus</i>	0.062	0.167	0.066	0.0	267.4	0.373	0.358
<i>Spratelloides gracilis</i>	0.042	0.108	0.110	0.0	78.3	0.384	0.380
<i>Enneapterygius</i>	0.018	0.035	0.129	13.1	27.9	0.520	0.167
<i>Cirrhilabrus ryukyuensis</i>	0.018	0.023	0.148	0.0	27.3	0.782	0.334
<i>Cirrhilabrus</i>	0.017	0.034	0.167	23.0	0.0	0.513	0.138
<i>Helcogramma desa</i>	0.017	0.036	0.184	40.5	3.8	0.463	0.147
<i>Spratelloides delicatulus</i>	0.015	0.035	0.200	0.0	31.5	0.432	0.631